
MA 465: Ordinary Differential Equations

Course Code	MA 465
Course Name	Ordinary Differential Equations
L-T-P-C	3-0-0-3
Prerequisites	MA 101
Intended for	B. Tech
Distribution	Elective
Approval	2nd Adhoc Senate (May 2012)

Course Contents

- **Basic Theory** [18 Lectures]

Existence and uniqueness of solutions, continuation of solutions, global existence, dependence of solutions on initial conditions, regularity of the flow, First and second order differential equations, Contraction mapping principle.

- **Linear Systems** [12 Lectures]

The fundamental matrix, Equilibrium points and their stability, Sturm-Liouville theory.

- **Nonlinear Systems** [10 Lectures]

The Poincare-Bendixon theorem, Perturbed systems, Lyapunov functional, Local and global analysis.

Text Books

1. Arnold, V., **Ordinary Differential Equations**, MIT Press, 1978.
2. Coddington, E. A. and Levinson, N., **Theory of Ordinary Differential Equations**, Krieger Publishing Co., 1984.
3. Ahmad, S. Rao, M.R.M., **Theory of ordinary differential equations with applications in biology and engineering**, EWP publication, 1999.

References

1. Perko, **Differential Equations and Dynamical Systems**, Springer.
2. Devaney, R., Hirsch, M. W. and Smale, S., **Differential Equations, Dynamical Systems, and an Introduction to Chaos**, 2nd Edition, Academic Press, 2003.
3. Birkhoff, G. and Rota, G.-C., **Ordinary Differential Equations**, Wiley, 1989.